

Volume

1

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PRINCIPLES, PROTOCOLS, AND ARCHITECTURES

FOURTH EDITION



DOUGLAS E. COMER

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Foreword

This is the fourth edition of a landmark book, the book that signaled the coming of age of the Internet. Development of the protocols for the Internet started around 1974, and they had been in limited but real use starting in the early 80's, but as of 1987, there was still no good introduction to how they worked or how to code them. The standards documents for TCP, IP and the other protocols existed, of course, but the true truth — the collection of knowledge and wisdom necessary to implement a protocol stack and actually expect it to work — that was a mystery, known only to a small band of the initiated. That was not a good thing, and the initiated knew it. But it takes a lot of effort to pull all the right stuff together and write it down. We waited, knowing that a good book explaining TCP/IP would be an important step towards the broad acceptance of our protocols.

And Doug wrote the book.

We told jokes, waiting for the book. We looked to see how many books there were in mature fields, and speculated that the number of books was a metric of success. I actually went and looked to see how many books there were on “how to build a compiler” (a post-mature field by now, perhaps — time to count the books again). The compiler community was well off, and even “how to build a database” was available. But nothing on “how to build a TCP/IP.” And then we got our book.

Of course, knowing that back then this was a landmark book is not enough to make you buy it. Collectors might want to find the first edition, but that gives the true truth as of 12 years ago, a long time in Internet years. And that is why this is the fourth edition. A lot has changed over that time. We have learned a lot more, the field has grown up, whole new protocols have emerged, and Doug has rewritten the book three times. That is a measure both of how much and how fast the field changes, and how much work must go into keeping this book current. It has all the new stuff, and our best current knowledge about all the old stuff.

Other things have changed in 12 years. Not only has the Internet grown up, but some of our heroes have grown old, and some have died. The foreword to the first edition was written by Jon Postel, one of the true Internet pioneers, who died in the fall of 1998. Below, we have reprinted the foreword he wrote for the first edition. Much is the same, but much has changed. This is still a very readable book both for details on TCP/IP and for an introduction to communications protocols in general. But in 1987, Jon wrote “Computer communication systems and networks are currently separated and

fragmented. The goal of interconnection and internetworking, to have a single powerful computer communication network, is fundamental to the design of TCP/IP.” Only 12 years ago networks were fragmented; today the Internet unites the world. And TCP/IP is still the glue, at the core of the Internet, that makes all this work. And this is still the book to read to learn about it.

David Clark
Massachusetts Institute of Technology

December, 1999

Foreword To The First Edition

By The Late Jon Postel

In this book Professor Douglas Comer has provided a long sought overview and introduction to TCP/IP. There have been many requests for “the” article, report, or book to read to get started on understanding the TCP/IP protocols. At last, this book satisfies those requests. Writing an introduction to TCP/IP for the uninitiated is a very difficult task. While combining the explanation of the general principles of computer communication with the specific examples from the TCP/IP protocol suite, Doug Comer has provided a very readable book.

While this book is specifically about the TCP/IP protocol suite, it is a good book for learning about computer communications protocols in general. The principles of architecture, layering, multiplexing, encapsulation, addressing and address mapping, routing, and naming are quite similar in any protocol suite, though, of course, different in detail (See Chapters 3, 10, 17, and 18)[†]. Computer communication protocols do not do anything themselves. Like operating systems, they are in the service of applications processes. Processes are the active elements that request communication and are the ultimate senders and receivers of the data transmitted. The various layers of protocols are like the various layers in a computer operating system, especially the file system. Understanding protocol architecture is like understanding operating system architecture. In this book Doug Comer has taken the “bottom up” approach — starting with the physical networks and moving up in levels of abstraction to the applications.

Since application processes are the active elements using the communication supported by the protocols, TCP/IP is an “interprocess communication” (IPC) mechanism. While there are several experiments in progress with operating system style message passing and procedure call types of IPC based on IP, the focus in this book is on more traditional applications that use the UDP datagram or TCP logical connection forms of IPC (See Chapters 11, 12, 17, 18, and 19).

One of the key ideas inherent in TCP/IP and in the title of this book is “internet-working.” The power of a communication system is directly related to the number of entities in that system. The telephone network is very useful because (nearly) all of the

[†]Editor’s note: chapter numbers have changed since the first edition.

telephones are in (as it appears to the users) one network. Computer communication systems and networks are currently separated and fragmented. The goal of interconnection and internetworking, to have a single powerful computer communication network, is fundamental to the design of TCP/IP. Essential to internetworking is addressing (See Chapters 4, 5, and 6), and a universal protocol — the Internet Protocol (See Chapters 7, 8, and 9).

To have an internetwork the individual networks must be connected. The connecting devices are called gateways. Further, these gateways must have some procedures for forwarding data from one network to the next. The data is in the form of IP datagrams and the destination is specified by an IP address, but the gateway must make a routing decision based on the IP address and what it knows about the connectivity of the networks making up the Internet. The procedures for distributing the current connectivity information to the gateways are called routing algorithms, and these are currently the subject of much study and development (See Chapters 13, 14, 15, and 16).

Like all communication systems, the TCP/IP protocol suite is an unfinished system. It is evolving to meet changing requirements and new opportunities. Thus, this book is, in a sense, a snapshot of TCP/IP circa 1987. And, as Doug Comer points out, there are many loose ends (See Chapter 20).

Most chapters end with a few pointers to material “for further study.” Many of these refer to memos of the RFC series of notes. This series of notes is the result of a policy of making the working ideas and the protocol specifications developed by the TCP/IP research and development community widely available. This availability of the basic and detailed information about these protocols, and the availability of the early implementations of them, has had much to do with their current widespread use. This commitment to public documentation at this level of detail is unusual for a research effort, and has had significant benefits for the development of computer communication (See Appendix 3).

This book brings together information about the various parts of the TCP/IP architecture and protocols and makes it accessible. Its publication is a very significant milestone in the evolution of computer communications.

Jon Postel,
Internet Protocol Designer and
Deputy Internet Architect

December, 1987

Preface

The explosive growth of the Internet continues. When the third edition of this book was written five years ago, the Internet connected 4.8 million computers, up from 5,000 when the first edition was published. The Internet now reaches over 56 million computers, meaning that the 1995 Internet was only about 8% of its current size. During the early 1990s, those of us who were involved with the Internet marveled at how large an obscure research project had become. Now, it pervades almost every aspect of society.

TCP/IP has accommodated change well. The basic technology has survived nearly two decades of exponential growth and the associated increases in traffic. The protocols have worked over new high-speed network technologies, and the design has handled applications that could not be imagined in the original design. Of course, the entire protocol suite has not remained static. New protocols have been deployed, and new techniques have been developed to adapt existing protocols to new network technologies.

This edition contains updated information throughout the text as well as new material that describes technical advances and changes. For example, because classless addressing has become widely deployed, the description of IP forwarding examines techniques for classless lookup. In addition, the chapters on IP describe the Differentiated Services (DiffServe) scheme for classes of service as well as path MTU discovery and anonymous networks. The chapter on TCP describes Random Early Drop (RED). The chapter on exterior routing has been updated to use BGP as the primary example. The descriptions of protocols such as RIP, IGMP, SNMP, and IPv6 have been revised to incorporate new versions and recent changes. Finally, the chapter on security discusses IPsec.

Four new chapters contain detailed information about significant developments. Chapter 19 describes mobile IP — a technology that allows a computer to move from one network to another without changing its IP address. Chapter 20 considers two technologies used to interconnect private intranets and the global Internet: Virtual Private Network (VPN) and Network Address Translation (NAT). Each solves a slightly different problem; both are widely deployed. Chapter 28 covers the HTML and HTTP protocols that form the basis for the most significant Internet application: the world wide web. Chapter 29 focuses on an exciting new area: sending real-time data such as

voice and video over an IP network. The chapter examines the RTP protocol that allows a receiver to coordinate and play such data as well as the RSVP and COPS protocols that can be used to provide resource reservation, and describes the H.323 suite of protocols used for IP telephony.

The fourth edition retains the same general contents and overall organization as the third edition. The entire text focuses on the concept of internetworking in general and the TCP/IP internet technology in particular. Internetworking is a powerful abstraction that allows us to deal with the complexity of multiple underlying communication technologies. It hides the details of network hardware and provides a high level communication environment. The text reviews both the architecture of network interconnections and the principles underlying protocols that make such interconnected networks function as a single, unified communication system. It also shows how an internet communication system can be used for distributed computation.

After reading this book, you will understand how it is possible to interconnect multiple physical networks into a coordinated system, how internet protocols operate in that environment, and how application programs use the resulting system. As a specific example, you will learn the details of the global TCP/IP Internet, including the architecture of its router system and the application protocols it supports. In addition, you will understand some of the limitations of the internet approach.

Designed as both a college text and as a professional reference, the book is written at an advanced undergraduate or graduate level. For professionals, the book provides a comprehensive introduction to the TCP/IP technology and the architecture of the Internet. Although it is not intended to replace protocol standards, the book is an excellent starting point for learning about internetworking because it provides a uniform overview that emphasizes principles. Moreover, it gives the reader perspective that can be extremely difficult to obtain from individual protocol documents.

When used in the classroom, the text provides more than sufficient material for a single semester network course at either the undergraduate or graduate level. Such a course can be extended to a two-semester sequence if accompanied by programming projects and readings from the literature. For undergraduate courses, many of the details are unnecessary. Students should be expected to grasp the basic concepts described in the text, and they should be able to describe or use them. At the graduate level, students should be expected to use the material as a basis for further exploration. They should understand the details well enough to answer exercises or solve problems that require them to explore extensions and subtleties. Many of the exercises suggest such subtleties; solving them often requires students to read protocol standards and apply creative energy to comprehend consequences.

At all levels, hands-on experience sharpens the concepts and helps students gain intuition. Thus, I encourage instructors to invent projects that force students to use Internet services and protocols. The semester project in my graduate Internetworking course at Purdue requires students to build an IP router. We supply hardware and the source code for an operating system, including device drivers for network interfaces; students build a working router that interconnects three networks with different MTUs. The course is extremely rigorous, students work in teams, and the results have been im-

pressive (many industries recruit graduates from the course). Although such experimentation is safest when the instructional laboratory network is isolated from production computing facilities, we have found that students exhibit the most enthusiasm, and benefit the most, when they have access to a functional TCP/IP internet.

The book is organized into four main parts. Chapters 1 and 2 form an introduction that provides an overview and discusses existing network technologies. In particular, Chapter 2 reviews physical network hardware. The intention is to provide basic intuition about what is possible, not to spend inordinate time on hardware details. Chapters 3-13 describe the TCP/IP Internet from the viewpoint of a single host, showing the protocols a host contains and how they operate. They cover the basics of Internet addressing and routing as well as the notion of protocol layering. Chapters 14-20 and 32 describe the architecture of an internet when viewed globally. They explore routing architecture and the protocols routers use to exchange routing information. Finally, Chapters 21-31 discuss application level services available in the Internet. They present the client-server model of interaction, and give several examples of client and server software.

The chapters have been organized bottom up. They begin with an overview of hardware and continue to build new functionality on top of it. This view will appeal to anyone who has developed Internet software because it follows the same pattern one uses in implementation. The concept of layering does not appear until Chapter 11. The discussion of layering emphasizes the distinction between conceptual layers of functionality and the reality of layered protocol software in which multiple objects appear at each layer.

A modest background is required to understand the material. The reader is expected to have a basic understanding of computer systems, and to be familiar with data structures like stacks, queues, and trees. Readers need basic intuition about the organization of computer software into an operating system that supports concurrent programming and application programs that users invoke to perform computation. Readers do not need sophisticated mathematics, nor do they need to know information theory or theorems from data communications; the book describes the physical network as a black box around which an internetwork can be built. It states design principles clearly, and discusses motivations and consequences.

I thank all the people who have contributed to versions of this book. Michael Evangelista provided extensive assistance with this edition, including classifying RFCs. Jeff Case provided the SNMPv3 example. John Lin and Dennis Totin commented on some of the new chapters. Jin Zhang, Kechiun He, and Sara Steinbrueck proofread parts of the text. Special thanks go to my wife and partner, Chris, whose careful editing made many improvements throughout.

Douglas E. Comer

January, 2000

What Others Have Said About The Fourth Edition Of Internetworking With TCP/IP

“This is the book I go to for clear explanations of the basic principles and latest developments in TCP/IP technologies. It’s a ‘must have’ reference for networking professionals.”

*Dr. Ralph Droms
Professor at Bucknell University*

“When the Nobel committee turns its attention to the Internet, Doug gets the prize for literature. This is an updated classic that is the best way to master Internet technology.”

*Dr. Paul V. Mockapetris
Inventor of the Domain Name System*

“The best-written TCP/IP book I have ever read. Dr. Comer explains complex ideas clearly, with excellent diagrams and explanations.”

*Dr. John Lin,
Bell Laboratories*

“Comer continues to prove himself the Baedeker of the Internet Protocols with this fine 4th edition.”

*Dr. Vinton Cerf
Senior Vice president, MCI WorldCom*

“There are many TCP/IP books on the shelves today, but Doug Comer’s ‘Internetworking with TCP/IP’ is the one that comes *off* the shelf for accessible and authoritative answers to questions about Internet technology.”

*Dr. Lyman Chapin,
Chief Scientist, BBN Technologies*

Other Books In the Internetworking Series from Douglas Comer and Prentice Hall

Internetworking With TCP/IP Volume II: Design, Implementation, and Internals (with David Stevens), 3rd edition: 1999, ISBN 0-13-973843-6

Volume II continues the discussion of Volume I by using code from a running implementation of TCP/IP to illustrate all the details. The text shows, for example, how TCP's slow start algorithm interacts with the Partridge-Karn exponential retransmission backoff algorithm and how routing updates interact with datagram forwarding.

Internetworking With TCP/IP Volume III: Client-Server Programming and Applications (with David Stevens)

BSD Socket Version, 2nd edition: 1996, ISBN 0-13-260969-X

AT&T TLI Version: 1994, ISBN 0-13-474230-3

Windows Sockets Version: 1997, ISBN 0-13-848714-6

Volume III describes the fundamental concept of client-server computing used to build all distributed computing systems. The text discusses various server designs as well as the tools and techniques used to build clients and servers, including Remote Procedure Call (RPC). It contains examples of running programs that illustrate each of the designs and tools. Three versions of Volume III are available for the socket API (Unix), the TLI API (AT&T System V), and the Windows Sockets API (Microsoft).

Computer Networks And Internets (with a CD-ROM by Ralph Droms), 2nd edition: 1999, ISBN 0-13-083617-6

A broad introduction to data communication, networking, internetworking, and client-server applications, *Computer Networks And Internets* examines the hardware and software components that make up computer networks, from the lowest levels through applications. The text covers transmission and modems, LANs and LAN extensions, access technologies, WANs, protocols (including TCP/IP), and network applications. The CD-ROM features animations and data sets.

The Internet Book: Everything you need to know about computer networking and how the Internet works, 2nd edition: 1997, ISBN 0-13-890161-9, paperback

A gentle introduction to networking and the Internet, *The Internet Book* does not assume the reader has a technical background. It explains the Internet, how it works, and services available in general terms, without focusing on a particular computer or a particular brand of software. Ideal for someone who wants to become Internet and computer networking literate, *The Internet Book* explains the terminology as well as the concepts; an extensive glossary of terms and abbreviations is included.

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Introduction And Overview

1.1 The Motivation For Internetworking

Internet communication has become a fundamental part of life. The World Wide Web contains information about such diverse subjects as atmospheric conditions, crop production, stock prices, and airline traffic. Groups establish electronic mailing lists so they can share information of common interest. Professional colleagues exchange business correspondence electronically, and relatives exchange personal greetings.

Unfortunately, most network technologies are designed for a specific purpose. Each enterprise chooses hardware technology appropriate for specific communication needs and budget. More important, it is impossible to engineer a universal network from a single network technology because no single network suffices for all uses. Some groups need high-speed networks to connect computers in a single building. Low-cost technologies that fill the need cannot span large geographic distances. Other groups settle for a slower speed network that connects machines thousands of miles apart.

For over two decades, a new technology has evolved that makes it possible to interconnect many disparate physical networks and make them function as a coordinated unit. The technology, called *internetworking*, accommodates multiple, diverse underlying hardware technologies by providing a way to interconnect heterogeneous networks and a set of communication conventions that makes them interoperate. The internet technology hides the details of network hardware, and permits computers to communicate independent of their physical network connections.

The internet technology described in this book is an example of *open system interconnection*. It is called *open* because, unlike proprietary communication systems available from one specific vendor, the specifications are publicly available. Thus, anyone can build the software needed to communicate across an internet. More important, the entire technology has been designed to foster communication among machines with

diverse hardware architectures, to use almost any packet switched network hardware, to accommodate a wide variety of applications, and to accommodate multiple computer operating systems.

To appreciate internet technology, think of how it has changed business. In addition to high-speed communication among employees in the office environment, networking technologies provide instant feedback among the production side of the business, sales and marketing, and customers. As a result, the speed with which business can plan, implement, assess, and retool has increased; the change is dramatic.

1.2 The TCP/IP Internet

U.S. government agencies realized the importance and potential of internet technology many years ago, and have funded research that has made possible a global Internet. This book discusses principles and ideas underlying the internet technology that has resulted from research funded by the *Advanced Research Projects Agency (ARPA)*[†]. The ARPA technology includes a set of network standards that specify the details of how computers communicate, as well as a set of conventions for interconnecting networks and routing traffic. Officially named the TCP/IP Internet Protocol Suite and commonly referred to as *TCP/IP* (after the names of its two main standards), it can be used to communicate across any set of interconnected networks. For example, some corporations use TCP/IP to interconnect all networks within their corporation, even though the corporation has no connection to outside networks. Other groups use TCP/IP for communication among geographically distant sites.

Although the TCP/IP technology is noteworthy by itself, it is especially interesting because its viability has been demonstrated on a large scale. It forms the base technology for the global Internet that connects over 170 million individuals in homes, schools, corporations, and government labs in virtually all populated countries. In the U.S., The *National Science Foundation (NSF)*, the *Department of Energy (DOE)*, the *Department of Defense (DOD)*, the *Health and Human Services Agency (HHS)*, and the *National Aeronautics and Space Administration (NASA)* have all participated in funding the Internet, and use TCP/IP to connect many of their research sites. Known as the *ARPA/NSF Internet*, the *TCP/IP Internet*, the *global Internet*, or just the *Internet*[‡], the resulting communication system allows subscribers to share information with anyone around the world as easily as they share it with someone in the next room. An outstanding success, the Internet demonstrates the viability of the TCP/IP technology and shows how it can accommodate a wide variety of underlying network technologies.

Most of the material in this book applies to any internet that uses TCP/IP, but some chapters refer specifically to the global Internet. Readers interested only in the technology should be careful to watch for the distinction between the Internet architecture as it exists and general TCP/IP internets as they might exist. It would be a mistake, however, to ignore all sections of the text that describe the global Internet — many corporate networks are already more complex than the global Internet of a dozen

[†]At various times, ARPA was called the *Defense Advanced Research Projects Agency (DARPA)*.

[‡]We will follow the usual convention of capitalizing *Internet* when referring specifically to the global Internet, and use lower case to refer to private internets that use TCP/IP technology.

years ago, and many of the problems they face have already been solved in the global Internet.

1.3 Internet Services

One cannot appreciate the technical details underlying TCP/IP without understanding the services it provides. This section reviews internet services briefly, highlighting the services most users access, and leaves to later chapters the discussion of how computers connect to a TCP/IP internet and how the functionality is implemented.

Much of our discussion of services will focus on standards called *protocols*. Protocols like TCP and IP provide the syntactic and semantic rules for communication. They contain the details of message formats, describe how a computer responds when a message arrives, and specify how a computer handles errors or other abnormal conditions. Most important, they allow us to discuss computer communication independent of any particular vendor's network hardware. In a sense, protocols are to communication what algorithms are to computation. An algorithm allows one to specify or understand a computation without knowing the details of a particular CPU instruction set. Similarly, a communication protocol allows one to specify or understand data communication without depending on detailed knowledge of a particular vendor's network hardware.

Hiding the low-level details of communication helps improve productivity in several ways. First, because programmers deal with higher-level protocol abstractions, they do not need to learn or remember as many details about a given hardware configuration. Thus, they can create new programs quickly. Second, because programs built using higher-level abstractions are not restricted to a particular computer architecture or a particular network hardware, they do not need to be changed when computers or networks are replaced or reconfigured. Third, because application programs built using higher-level protocols are independent of the underlying hardware, they can provide direct communication between an arbitrary pair of computers. Programmers do not need to build a special version of application software for each type of computer or each type of network. Instead, software built to use protocols is general-purpose; the same code can be compiled and run on an arbitrary computer.

We will see that the details of each service available on the Internet are given by a separate protocol. The next sections refer to protocols that specify some of the application-level services as well as those used to define network-level services. Later chapters explain each of these protocols in detail.

1.3.1 Application Level Internet Services

From the user's point of view, the Internet appears to consist of a set of application programs that use the underlying network to carry out useful communication tasks. We use the term *interoperability* to refer to the ability of diverse computing systems to cooperate in solving computational problems. Internet application programs exhibit a high degree of interoperability. Most users that access the Internet do so merely by run-

ning application programs without understanding the types of computers being accessed, the TCP/IP technology, the structure of the underlying internet, or even the path the data travels to its destination; they rely on the application programs and the underlying network software to handle such details. Only programmers who write network application programs need to view a TCP/IP internet as a network and need to understand some of the technology.

The most popular and widespread Internet application services include:

- *World Wide Web.* The Web allows users to view documents that contain text and graphics, and to follow hypermedia links from one document to another. The Web grew to become the largest source of traffic on the global Internet between 1994 and 1995, and continues to dominate. Some service providers estimate that the Web now accounts for 80% of their Internet traffic.
- *Electronic mail (e-mail).* Electronic mail allows a user to compose a memo and send a copy to individuals or groups. Another part of the mail application allows users to read memos that they have received. A recent innovation allows users to include “attachments” with a mail message that consist of arbitrary files. Electronic mail has been so successful that many Internet users depend on it for most correspondence. One reason for the popularity of Internet e-mail arises from a careful design: the protocol makes delivery reliable. Not only does the mail system on the sender’s computer contact the mail system on the receiver’s computer directly, but the protocol specifies that a message cannot be deleted by the sender until the receiver has successfully placed a copy on permanent storage.
- *File transfer.* The file transfer application allows users to send or receive a copy of a data file. File transfer is one of the oldest, and still among the most heavily used application services in the Internet. Although small files can now be attached to an e-mail message, the file transfer service is still needed to handle arbitrarily large files. The system provides a way to check for authorized users, or even to prevent all access. Like mail, file transfer across a TCP/IP internet is reliable because the two machines involved communicate directly, without relying on intermediate machines to make copies of the file along the way.
- *Remote login.* Remote login allows a user sitting at one computer to connect to a remote machine and establish an interactive login session. The remote login makes it appear that a window on the user’s screen connects directly to the remote machine by sending each keystroke from the user’s keyboard to the remote machine and displaying each character the remote computer prints in the user’s window. When the remote login session terminates, the application returns the user to the local system.

We will return to these and other applications in later chapters to examine them in more detail. We will see exactly how they use the underlying TCP/IP protocols, and why having standards for application protocols has helped ensure that they are widespread.

1.3.2 Network-Level Internet Services

A programmer who creates application programs that use TCP/IP protocols has an entirely different view of an internet than a user who merely executes applications like electronic mail. At the network level, an internet provides two broad types of service that all application programs use. While it is unimportant at this time to understand the details of these services, they cannot be omitted from any overview of TCP/IP:

- *Connectionless Packet Delivery Service.* This service, explained in detail throughout the text, forms the basis for all other internet services. Connectionless delivery is an abstraction of the service that most packet-switching networks offer. It means simply that a TCP/IP internet routes small messages from one computer to another based on address information carried in the message. Because the connectionless service routes each packet separately, it does not guarantee reliable, in-order delivery. Because it usually maps directly onto the underlying hardware, the connectionless service is extremely efficient. More important, having connectionless packet delivery as the basis for all internet services makes the TCP/IP protocols adaptable to a wide range of network hardware.
- *Reliable Stream Transport Service.* Most applications need much more than packet delivery because they require the communication software to recover automatically from transmission errors, lost packets, or failures of intermediate switches along the path between sender and receiver. The reliable transport service handles such problems. It allows an application on one computer to establish a “connection” with an application on another computer, and then to send a large volume of data across the connection as if it were a permanent, direct hardware connection. Underneath, of course, the communication protocols divide the stream of data into small messages and send them, one at a time, waiting for the receiver to acknowledge reception.

Many networks provide basic services similar to those outlined above, so one might wonder what distinguishes TCP/IP services from others. The primary distinguishing features are:

- *Network Technology Independence.* Although TCP/IP is based on conventional packet switching technology, it is independent of any particular vendor’s hardware. The global Internet includes a variety of network technologies ranging from networks designed to operate within a single building to those designed to span large distances. TCP/IP protocols define the unit of data transmission, called a *datagram*, and specify how to transmit datagrams on a particular network.
- *Universal Interconnection.* A TCP/IP internet allows any pair of computers to which it attaches to communicate. Each computer is assigned an *address* that is universally recognized throughout the internet. Every datagram carries the addresses of its source and destination. Intermediate switching computers use the destination address to make routing decisions.

- *End-to-End Acknowledgements.* The TCP/IP internet protocols provide acknowledgements between the original source and ultimate destination instead of between successive machines along the path, even if the source and destination do not connect to a common physical network.
- *Application Protocol Standards.* In addition to the basic transport-level services (like reliable stream connections), the TCP/IP protocols include standards for many common applications including electronic mail, file transfer, and remote login. Thus, when designing application programs that use TCP/IP, programmers often find that existing software provides the communication services they need.

Later chapters will discuss the details of the services provided to the programmer as well as many of the application protocol standards.

1.4 History And Scope Of The Internet

Part of what makes the TCP/IP technology so exciting is its universal adoption as well as the size and growth rate of the global Internet. ARPA began working toward an internet technology in the mid 1970s, with the architecture and protocols taking their current form around 1977-79. At that time, ARPA was known as the primary funding agency for packet-switched network research and had pioneered many ideas in packet-switching with its well-known *ARPANET*. The ARPANET used conventional point-to-point leased line interconnection, but ARPA had also funded exploration of packet-switching over radio networks and satellite communication channels. Indeed, the growing diversity of network hardware technologies helped force ARPA to study network interconnection, and pushed internetworking forward.

The availability of research funding from ARPA caught the attention and imagination of several research groups, especially those researchers who had previous experience using packet switching on the ARPANET. ARPA scheduled informal meetings of researchers to share ideas and discuss results of experiments. Informally, the group was known as the *Internet Research Group*. By 1979, so many researchers were involved in the TCP/IP effort that ARPA created an informal committee to coordinate and guide the design of the protocols and architecture of the emerging Internet. Called the Internet Control and Configuration Board (*ICCB*), the group met regularly until 1983, when it was reorganized.

The global Internet began around 1980 when ARPA started converting machines attached to its research networks to the new TCP/IP protocols. The ARPANET, already in place, quickly became the backbone of the new Internet and was used for many of the early experiments with TCP/IP. The transition to Internet technology became complete in January 1983 when the Office of the Secretary of Defense mandated that all computers connected to long-haul networks use TCP/IP. At the same time, the *Defense Communication Agency (DCA)* split the ARPANET into two separate networks, one for further research and one for military communication. The research part retained the name ARPANET; the military part, which was somewhat larger, became known as the *military network, MILNET*.

To encourage university researchers to adopt and use the new protocols, ARPA made an implementation available at low cost. At that time, most university computer science departments were running a version of the UNIX operating system available in the University of California's *Berkeley Software Distribution*, commonly called *Berkeley UNIX* or *BSD UNIX*. By funding Bolt Beranek and Newman, Incorporated (*BBN*) to implement its TCP/IP protocols for use with UNIX and funding Berkeley to integrate the protocols with its software distribution, ARPA was able to reach over 90% of university computer science departments. The new protocol software came at a particularly significant time because many departments were just acquiring second or third computers and connecting them together with local area networks. The departments needed communication protocols.

The Berkeley software distribution became popular because it offered more than basic TCP/IP protocols. In addition to standard TCP/IP application programs, Berkeley offered a set of utilities for network services that resembled the UNIX services used on a single machine. The chief advantage of the Berkeley utilities lies in their similarity to standard UNIX. For example, an experienced UNIX user can quickly learn how to use Berkeley's remote file copy utility (*rcp*) because it behaves exactly like the UNIX file copy utility except that it allows users to copy files to or from remote machines.

Besides a set of utility programs, Berkeley UNIX provided a new operating system abstraction known as a *socket* that allowed application programs to access communication protocols. A generalization of the UNIX mechanism for I/O, the socket has options for several types of network protocols in addition to TCP/IP. Its design has been debated since its introduction, and many operating systems researchers have proposed alternatives. Independent of its overall merits, however, the introduction of the socket abstraction was important because it allowed programmers to use TCP/IP protocols with little effort. Thus, it encouraged researchers to experiment with TCP/IP.

The success of the TCP/IP technology and the Internet among computer science researchers led other groups to adopt it. Realizing that network communication would soon be a crucial part of scientific research, the National Science Foundation (*NSF*) took an active role in expanding the TCP/IP Internet to reach as many scientists as possible. In the late 1970s, NSF funded a project known as the *Computer Science Network (CSNET)*, which had as its goal connecting all computer scientists. Starting in 1985, NSF began a program to establish access networks centered around its six supercomputer centers. In 1986 it expanded networking efforts by funding a new wide area backbone network, called the *NSFNET*[†], that eventually reached all its supercomputer centers and tied them to the ARPANET. Finally, in 1986 NSF provided seed money for many regional networks, each of which now connects major scientific research institutions in a given area. All the NSF-funded networks use TCP/IP protocols, and all are part of the global Internet.

Within seven years of its inception, the Internet had grown to span hundreds of individual networks located throughout the United States and Europe. It connected nearly 20,000 computers at universities, government, and corporate research laboratories. Both the size and the use of the Internet continued to grow much faster than anticipated. By

[†]The term *NSFNET* is sometimes used loosely to mean all NSF-funded networking activities, but we will use it to refer to the backbone. The next chapter gives more details about the technology.

late 1987, it was estimated that the growth had reached 15% per month. By 2000, the global Internet reached over 50 million computers in 209 countries.

Early adoption of TCP/IP protocols and growth of the Internet has not been limited to government-funded projects. Major computer corporations connected to the Internet as did many other large corporations including: oil companies, the auto industry, electronics firms, pharmaceutical companies, and telecommunications carriers. Medium and small companies began connecting in the 1990s. In addition, many companies have used the TCP/IP protocols on their internal corporate internets even though they choose not to be part of the global Internet.

Rapid expansion introduced problems of scale unanticipated in the original design and motivated researchers to find techniques for managing large, distributed resources. In the original design, for example, the names and addresses of all computers attached to the Internet were kept in a single file that was edited by hand and then distributed to every site on the Internet. By the mid 1980s, it became apparent that a central database would not suffice. First, because computers were being added to the Internet at an increasing rate, requests to update the file would soon exceed the personnel available to process them. Second, even if a correct central file existed, network capacity was insufficient to allow either frequent distribution to every site or on-line access by each site.

New protocols were developed and a naming system was put in place across the global Internet that allows any user to resolve the name of a remote machine automatically. Known as the *Domain Name System (DNS)*, the mechanism relies on machines called *name servers* to answer queries about names. No single machine contains the entire domain name database. Instead, data is distributed among a set of machines that use TCP/IP protocols to communicate among themselves when answering a query.

1.5 The Internet Architecture Board

Because the TCP/IP internet protocol suite did not arise from a specific vendor or from a recognized professional society, it is natural to ask, “who sets the technical direction and decides when protocols become standard?” The answer is a group known as the *Internet Architecture Board (IAB)*[†]. The IAB provides the focus and coordination for much of the research and development underlying the TCP/IP protocols, and guides the evolution of the Internet. It decides which protocols are a required part of the TCP/IP suite and sets official policies.

Formed in 1983 when ARPA reorganized the Internet Control and Configuration Board, the IAB inherited much of its charter from the earlier group. Its initial goals were to encourage the exchange of ideas among the principals involved in research related to TCP/IP and the Internet, and to keep researchers focused on common objectives. Through the first six years, the IAB evolved from an ARPA-specific research group into an autonomous organization. During these years, each member of the IAB chaired an *Internet Task Force* charged with investigating a problem or set of issues deemed to be important. The IAB consisted of approximately ten task forces, with charters ranging from one that investigated how the traffic load from various applica-

[†]IAB originally stood for *Internet Activities Board*.

tions affects the Internet to one that handled short term Internet engineering problems. The IAB met several times each year to hear status reports from each task force, review and revise technical directions, discuss policies, and exchange information with representatives from agencies like ARPA and NSF, who funded Internet operations and research.

The chairman of the IAB had the title *Internet Architect* and was responsible for suggesting technical directions and coordinating the activities of the various task forces. The IAB chairman established new task forces on the advice of the IAB and also represented the IAB to others.

Newcomers to TCP/IP are sometimes surprised to learn that the IAB did not manage a large budget; although it set direction, it did not fund most of the research and engineering it envisioned. Instead, volunteers performed much of the work. Members of the IAB were each responsible for recruiting volunteers to serve on their task forces, for calling and running task force meetings, and for reporting progress to the IAB. Usually, volunteers came from the research community or from commercial organizations that produced or used TCP/IP. Active researchers participated in Internet task force activities for two reasons. On one hand, serving on a task force provided opportunities to learn about new research problems. On the other hand, because new ideas and problem solutions designed and tested by task forces often became part of the TCP/IP Internet technology, members realized that their work had a direct, positive influence on the field.

1.6 The IAB Reorganization

By the summer of 1989, both the TCP/IP technology and the Internet had grown beyond the initial research project into production facilities on which thousands of people depended for daily business. It was no longer possible to introduce new ideas by changing a few installations overnight. To a large extent, the literally hundreds of commercial companies that offer TCP/IP products determined whether products would interoperate by deciding when to incorporate changes in their software. Researchers who drafted specifications and tested new ideas in laboratories could no longer expect instant acceptance and use of the ideas. It was ironic that the researchers who designed and watched TCP/IP develop found themselves overcome by the commercial success of their brainchild. In short, TCP/IP became a successful, production technology and the market place began to dominate its evolution.

To reflect the political and commercial realities of both TCP/IP and the Internet, the IAB was reorganized in the summer of 1989. The chairmanship changed. Researchers were moved from the IAB itself to a subsidiary group and a new IAB board was constituted to include representatives from the wider community.

Figure 1.1 illustrates the IAB organization and the relationship of subgroups.

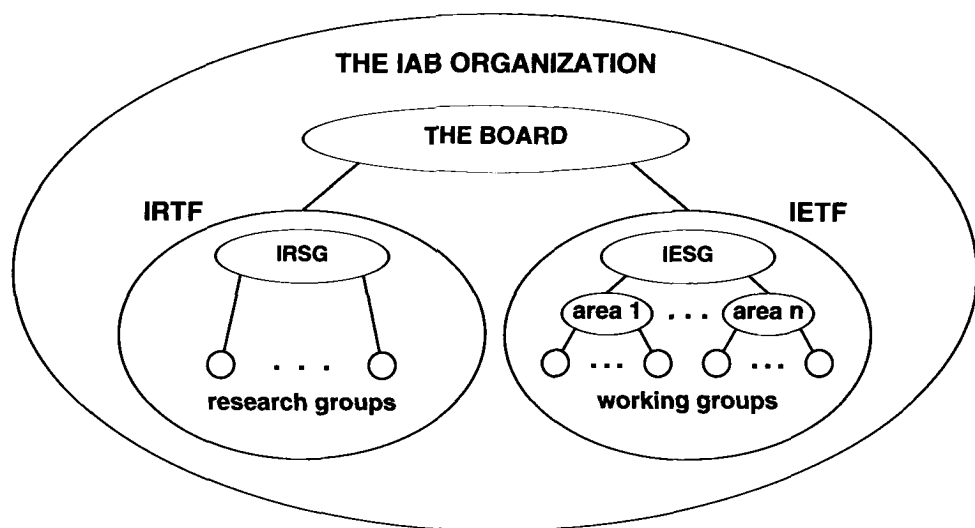


Figure 1.1 The structure of the IAB after the 1989 reorganization.

As Figure 1.1 shows, in addition to the board itself, the IAB organization contained two major groups: the *Internet Research Task Force (IRTF)* and the *Internet Engineering Task Force (IETF)*.

As its name implies, the IETF concentrates on short-term or medium-term engineering problems. The IETF existed in the original IAB structure, and its success provided part of the motivation for reorganization. Unlike most IAB task forces, which were limited to a few individuals who focused on one specific issue, the IETF was large — before the reorganization, it had grown to include dozens of active members who worked on many problems concurrently. It was divided into over 20 *working groups*, each focusing on a specific problem. Working groups held individual meetings to formulate problem solutions. In addition, the entire IETF met regularly to hear reports from working groups and discuss proposed changes or additions to the TCP/IP technology. Usually held three times annually, full IETF meetings attracted hundreds of participants and spectators. The IETF had become too large for the chairman to manage.

Because the IETF was known throughout the Internet, and because its meetings were widely recognized and attended, the reorganized IAB structure retains the IETF, but splits it into approximately ten areas, each with its own manager. The IETF chairman and the area managers comprise the *Internet Engineering Steering Group (IESG)*, the individuals responsible for coordinating the efforts of IETF working groups. The name “IETF” now refers to the entire body, including the chairman, area managers, and all members of working groups.

Created during the reorganization, the Internet Research Task Force is the research counterpart to the IETF. The IRTF coordinates research activities related to TCP/IP protocols or internet architecture in general. Like the IETF, the IRTF has a small group, called the *Internet Research Steering Group (IRSG)*, that sets priorities and coordinates research activities. Unlike the IETF, the IRTF is currently a much smaller and less active organization. In fact, most of the research is being done within the IETF.

1.7 The Internet Society

In 1992, as the Internet moved away from its U.S. government roots, a society was formed to encourage participation in the Internet. Called the *Internet Society (ISOC)*, the group is an international organization inspired by the National Geographic Society. The host for the IAB, the Internet Society continues to help people join and use the Internet around the world.

1.8 Internet Request For Comments

We have said that no vendor owns the TCP/IP technology nor does any professional society or standards body. Thus, the documentation of protocols, standards, and policies cannot be obtained from a vendor. Instead, the documentation is placed in on-line repositories and made available at no charge.

Documentation of work on the Internet, proposals for new or revised protocols, and TCP/IP protocol standards all appear in a series of technical reports called *Internet Requests For Comments*, or *RFCs*. RFCs can be short or long, can cover broad concepts or details, and can be standards or merely proposals for new protocols[†]. While RFCs are not refereed in the same way as academic research papers, they are edited. For many years, a single individual, Jon Postel[‡], served as RFC editor. The task of editing RFCs now falls to area managers of the IETF; the IESG as a whole approves new RFCs.

Finally, a few reports pertinent to the Internet were published in an earlier, parallel series of reports called *Internet Engineering Notes*, or *IENs*. Although the IEN series is no longer active, not all IENs appear in the RFC series. There are references to RFCs (and still a few to IENs) throughout the text.

The RFC series is numbered sequentially in the chronological order RFCs are written. Each new or revised RFC is assigned a new number, so readers must be careful to obtain the highest numbered version of a document; an RFC index is available to help identify the correct version.

To make document retrieval quicker, many sites around the world store copies of RFCs and make them available to the community. One can obtain RFCs by postal mail, by electronic mail, or directly across the Internet using a file transfer program. In addition, preliminary versions of RFC documents, which are known as *Internet drafts*,

[†]Appendix I contains an introduction to RFCs that examines the diversity of RFCs, including jokes that have appeared.

[‡]Jon passed away in the fall of 1998. He was one of the pioneers who made significant contributions to TCP/IP and the Internet. Those of us who knew him feel the loss deeply.

are also available. Ask a local network expert how to obtain RFCs or Internet drafts at your site, or refer to Appendix 1 for further instructions on how to retrieve them.

1.9 Internet Protocols And Standardization

Readers familiar with data communication networks realize that a myriad of communication protocol standards exist. Many of them precede the Internet, so the question arises, “Why did the Internet designers invent new protocols when so many international standards already existed?” The answer is complex, but follows a simple maxim:

Use existing protocol standards whenever such standards apply; invent new protocols only when existing standards are insufficient, and be prepared to use new standards when they become available and provide equivalent functionality.

So, despite appearances to the contrary, the TCP/IP Internet Protocol Suite was not intended to ignore or avoid extant standards. It came about merely because none of the existing protocols satisfied the need for an interoperable internetworking communication system.

1.10 Future Growth And Technology

Both the TCP/IP technology and the Internet continue to evolve. New protocols are being proposed; old ones are being revised. NSF added considerable complexity to the system by introducing a backbone network, regional networks, and hundreds of campus networks. Other groups around the world continue to connect to the Internet as well. The most significant change comes not from added network connections, however, but from additional traffic. As new users connect to the Internet and new applications appear, traffic patterns change. When physicists, chemists, and biologists began to use the Internet, they exchanged files of data collected from experiments. Files of scientific data were large compared to electronic mail messages. As the Internet became popular and users began to browse information using services like the *World Wide Web*, traffic patterns increased again.

To accommodate growth in traffic, the capacity of the NSFNET backbone was increased three times. The final version, known as *ANSNET* after the company that supplied it, had a capacity approximately 840 times larger than the original. Since 1995, companies known as *Internet Service Providers (ISPs)* have each built their own backbone network, many of which have significantly more capacity than the last government-funded backbone. At the current time, it is difficult to foresee an end to the need for more capacity.

Growth in demands for networking is not unexpected. The computer industry has enjoyed a continual demand for increased processing power and larger data storage for many years. Users have only begun to understand how to use networks. In the future we can expect continual increases in the demand for communications. Soon, for example, TCP/IP technologies will be used for telephone and video services as well as data services. Thus, higher-capacity communication technologies will be needed to accommodate the growth.

Figure 1.2 summarizes expansion of the Internet and illustrates an important component of growth: much of the change in complexity has arisen because multiple groups now manage various parts of the whole. Because the technology was developed when a single person at ARPA had control of all aspects of the Internet, the designs of many subsystems depended on centralized management and control. As the Internet grew, responsibility and control were divided among multiple organizations. In particular, as the Internet became global, the operation and management needed to span multiple countries. Much of the effort since the early 1990s has been directed toward finding ways to extend the design to accommodate decentralized management.

	number of networks	number of computers	number of users	number of managers
1980	10	10 ²	10 ²	10 ⁰
1990	10 ³	10 ⁵	10 ⁶	10 ¹
2000	10 ⁵	10 ⁷	10 ⁸	10 ²

Figure 1.2 Growth of the connected Internet. In addition to traffic increases that result from increased size, the Internet faces complexity that results from decentralized management of both development and operations.

1.11 Organization Of The Text

The material on TCP/IP has been written in three volumes. This volume presents the TCP/IP technology, applications that use it, and the architecture of the global Internet in more detail. It discusses the fundamentals of protocols like TCP and IP, and shows how they fit together in an internet. In addition to giving details, the text highlights the general principles underlying network protocols, and explains why the TCP/IP protocols adapt easily to so many underlying physical network technologies. Volume II discusses in depth the internal details of the TCP/IP protocols and shows how they are implemented. It presents code from a working system to illustrate how the individual protocols work together, and contains details useful to people responsible

for building a corporate internet. Volume III shows how distributed applications use TCP/IP for communication. It focuses on the client-server paradigm, the basis for all distributed programming. It discusses the interface between programs and protocols†, and shows how client and server programs are organized. In addition, Volume III describes the remote procedure concept, middleware, and shows how programmers use tools to build client and server software.

So far, we have talked about the TCP/IP technology and the Internet in general terms, summarizing the services provided and the history of their development. The next chapter provides a brief summary of the type of network hardware used throughout the Internet. Its purpose is not to illuminate nuances of a particular vendor's hardware, but to focus on the features of each technology that are of primary importance to an internet architect. Later chapters delve into the protocols and the Internet, fulfilling three purposes: they explore general concepts and review the Internet architectural model, they examine the details of TCP/IP protocols, and they look at standards for high-level services like electronic mail and electronic file transfer. Chapters 3 through 14 review fundamental principles and describe the network protocol software found in any machine that uses TCP/IP. Later chapters describe services that span multiple machines, including the propagation of routing information, name resolution, and applications like electronic mail.

Two appendices follow the main text. The first appendix contains a guide to RFCs. It expands on the description of RFCs found in this chapter, and gives examples of information that can be found in RFCs. It describes in detail how to obtain RFCs by electronic mail, postal mail, and file transfer. Finally, because the standard RFC index comes in chronological order, the appendix presents a list of RFCs organized by topic to make it easier for beginners to find RFCs pertinent to a given subject.

The second appendix contains an alphabetical list of terms and abbreviations used throughout the literature and the text. Because beginners often find the new terminology overwhelming and difficult to remember, they are encouraged to use the alphabetical list instead of scanning back through the text.

1.12 Summary

An internet consists of a set of connected networks that act as a coordinated whole. The chief advantage of an internet is that it provides universal interconnection while allowing individual groups to use whatever network hardware is best suited to their needs. We will examine principles underlying internet communication in general and the details of one internet protocol suite in particular. We will also discuss how internet protocols are used in an internet. Our example technology, called TCP/IP after its two main protocols, was developed by the Advanced Research Projects Agency. It provides the basis for the global Internet, a large, operational internet that connects universities, corporations, and government departments in many countries around the world. The global Internet is expanding rapidly.

†Volume III is available in three versions: one that uses the Unix *socket interface* interface in examples, a second that uses the *Transport Layer Interface (TLI)*, and a third that uses the *Windows Sockets Interface* defined by Microsoft.

FOR FURTHER STUDY

Cerf's *A History Of The ARPANET* [1989] and *History of the Internet Activities Board* [RFC 1160] provide fascinating reading and point the reader to early research papers on TCP/IP and internetworking. Denning [Nov-Dec 1989] provides a different perspective on the history of the ARPANET. Jennings et. al. [1986] discusses the importance of computer networking for scientists. Denning [Sept-Oct 1989] also points out the importance of internetworking and gives one possible scenario for a world-wide Internet. The U.S. Federal Coordinating Committee for Science, Engineering and Technology [FCCSET] suggested networking should be a national priority.

The IETF (ietf.org) publishes minutes from its regular meetings. The Internet Society (www.isoc.org) produces newsletters that discuss the penetration of the Internet in countries around the world. The World Wide Web Consortium (w3c.org) produces protocols and standards for Web technologies. Finally, the reader is encouraged to remember that the TCP/IP protocol suite and the Internet continue to evolve; new information can be found in RFCs and at conferences such as the annual ACM SIGCOMM Symposium and NETWORLD+INTEROP events held around the world.

EXERCISES

- 1.1 Explore application programs at your site that use TCP/IP.
- 1.2 Plot the growth of TCP/IP technology and Internet access at your organization. How many computers, users, and networks were connected each year?
- 1.3 TCP/IP products account for several billion dollars per year in gross revenue. Read trade publications to find a list of vendors offering such products.

Review Of Underlying Network Technologies

2.1 Introduction

It is important to understand that the Internet is not a new kind of physical network. It is, instead, a method of interconnecting physical networks and a set of conventions for using networks that allow the computers they reach to interact. While network hardware plays only a minor role in the overall design, understanding the internet technology requires one to distinguish between the low-level mechanisms provided by the hardware itself and the higher-level facilities that the TCP/IP protocol software provides. It is also important to understand how the interfaces supplied by underlying packet-switched technology affect our choice of high-level abstractions.

This chapter introduces basic packet-switching concepts and terminology, and then reviews some of the underlying network hardware technologies that have been used in TCP/IP internets. Later chapters describe how these networks are interconnected and how the TCP/IP protocols accommodate vast differences in the hardware. While the list presented here is certainly not comprehensive, it clearly demonstrates the variety among physical networks over which TCP/IP operates. The reader can safely skip many of the technical details, but should try to grasp the idea of packet switching and try to imagine building a homogeneous communication system using such heterogeneous hardware. Most important, the reader should look closely at the details of the physical address schemes the various technologies use; later chapters will discuss in detail how high-level protocols use physical addresses.

2.2 Two Approaches To Network Communication

Whether they provide connections between one computer and another or between a terminal and a computer, communication networks can be divided into two basic types: *connection-oriented* (sometimes called *circuit-switched*) and *connectionless* (sometimes called *packet-switched*[†]). Connection-oriented networks operate by forming a dedicated *connection* or *circuit* between two points. The U.S. telephone system uses a connection-oriented technology — a telephone call establishes a connection from the originating phone through the local switching office, across trunk lines, to a remote switching office, and finally to the destination telephone. While a connection is in place, the phone equipment samples the microphone repeatedly, encodes the samples digitally, and transmits them across the connection to the receiver. The sender is guaranteed that the samples can be delivered and reproduced because the connection provides a guaranteed data path of 64 Kbps (thousand bits per second), the rate needed to send digitized voice. The advantage of connection-oriented networking lies in its guaranteed capacity: once a circuit is established, no other network activity will decrease the capacity of that circuit. One disadvantage of connection-oriented technology arises from cost: circuit costs are fixed, independent of use. For example, one pays a fixed rate for a phone call, even when the two parties do not talk.

Connectionless networks, the type often used to connect computers, take an entirely different approach. In a connectionless network, data to be transferred across a network is divided into small pieces called *packets* that are multiplexed onto high capacity intermachine connections. A packet, which usually contains only a few hundred bytes of data, carries identification that enables the network hardware to know how to send it to the specified destination. For example, a large file to be transmitted between two machines must be broken into many packets that are sent across the network one at a time. The network hardware delivers the packets to the specified destination, where software reassembles them into a single file again. The chief advantage of packet-switching is that multiple communications among computers can proceed concurrently, with intermachine connections shared by all pairs of computers that are communicating. The disadvantage, of course, is that as activity increases, a given pair of communicating computers receives less of the network capacity. That is, whenever a packet switched network becomes overloaded, computers using the network must wait before they can send additional packets.

Despite the potential drawback of not being able to guarantee network capacity, connectionless networks have become extremely popular. The motivations for adopting packet switching are cost and performance. Because multiple computers can share the network bandwidth, fewer connections are required and cost is kept low. Because engineers have been able to build high speed network hardware, capacity is not usually a problem. So many computer interconnections use connectionless networks that, throughout the remainder of this text, we will assume the term *network* refers to a connectionless network unless otherwise stated.

[†]In fact, it is possible to build hybrid hardware technologies; for our purposes, only the difference in functionality is important.

2.3 Wide Area And Local Area Networks

Data networks that span large geographical distances (e.g., the continental U.S.) are fundamentally different from those that span short distances (e.g., a single room). To help characterize the differences in capacity and intended use, packet switched technologies are often divided into two broad categories: *wide area networks* (WANs) and *Local Area Networks* (LANs). The two categories do not have formal definitions. Instead, vendors apply the terms loosely to help customers distinguish among technologies.

WAN technologies, sometimes called *long haul networks*, provide communication over long distances. Most WAN technologies do not limit the distance spanned; a WAN can allow the endpoints of a communication to be arbitrarily far apart. For example, a WAN can span a continent or can join computers across an ocean. Usually, WANs operate at slower speeds than LANs, and have much greater delay between connections. Typical speeds for a WAN range from 1.5 Mbps to 155 Mbps (million bits per second). Delays across a WAN can vary from a few milliseconds to several tenths of a second†.

LAN technologies provide the highest speed connections among computers, but sacrifice the ability to span long distances. For example, a typical LAN spans a small area like a single building or a small campus, and operates between 10 Mbps and 2 Gbps (billion bits per second). Because LAN technologies cover short distances, they offer lower delays than WANs. The delay across a LAN can be as short as a few tenths of a millisecond or as long as 10 milliseconds.

We have already stated the general tradeoff between speed and distance: technologies that provide higher speed communication operate over shorter distances. There are other differences among the technologies as well. In LAN technologies, each computer usually contains a device known as a *Network Interface Card (NIC)* that connects the machine directly to the network. The network itself need not contain much intelligence; it can depend on electronic interface devices in the attached computers to generate and receive the complex electrical signals. In WAN technologies, a network usually consists of a series of complex computers called *packet switches* interconnected by long-distance communication lines. The size of the network can be extended by adding a new switch and another communication line. Attaching a user's computer to a WAN means connecting it to one of the packet switches. Each switch along a path in the WAN introduces delay when it receives a packet and forwards it to the next switch. Thus, the larger the WAN becomes the longer it takes to route traffic across it.

This book discusses software that hides the technological differences among networks and makes interconnection independent of the underlying hardware. To appreciate design choices in the software, it is necessary to understand how it relates to network hardware. The next sections present examples of network technologies that have been used in the Internet, showing some of the differences among them. Later chapters show how the TCP/IP software isolates such differences and makes the communication system independent of the underlying hardware technology.

†Such long delays result from WANs that communicate by sending signals to a satellite orbiting the earth.

2.3.1 Network Hardware Addresses

Each network hardware technology defines an *addressing mechanism* that computers use to specify the destination for a packet. Every computer attached to a network is assigned a unique address, usually an integer. A packet sent across a network includes a *destination address field* that contains the address of the intended recipient. The destination address appears in the same location in all packets, making it possible for the network hardware to examine the destination address easily. A sender must know the address of the intended recipient, and must place the recipient's address in the destination address field of a packet before transmitting the packet.

Each hardware technology specifies how computers are assigned addresses. The hardware specifies, for example, the number of bits in the address as well as the location of the destination address field in a packet. Although some technologies use compatible addressing schemes, many do not. This chapter contains a few examples of hardware addressing schemes; later chapters explain how TCP/IP accommodates diverse hardware addressing schemes.

2.4 Ethernet Technology

Ethernet is the name given to a popular packet-switched LAN technology invented at Xerox PARC in the early 1970s. Xerox Corporation, Intel Corporation, and Digital Equipment Corporation standardized Ethernet in 1978; IEEE released a compatible version of the standard using the standard number 802.3. Ethernet has become the most popular LAN technology; it now appears in virtually all corporate networks as well as many small installations. Because Ethernet is so popular, many variants exist. Although the original wiring scheme has been superseded, understanding the original design helps clarify the intent and some of the design decisions. Thus, we will discuss the original design first, and then cover variants.

Formally known as *10Base5*, the original Ethernet design uses a coaxial cable as Figure 2.1 illustrates.

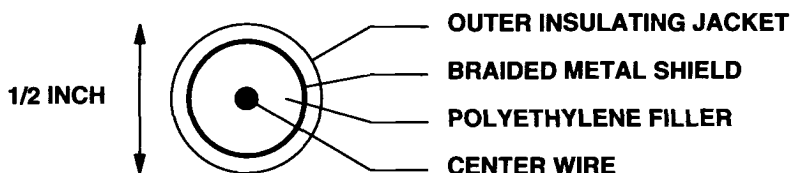


Figure 2.1 A cross-section of the coaxial cable used in the original Ethernet.

Called the *ether*, the cable itself is completely passive; all the active electronic components needed to make the network function are associated with the computers attached to the network. Each Ethernet cable is about 1/2 inch in diameter and up to 500

meters long. A resistor is added between the center wire and shield at each end to prevent reflection of electrical signals.

The connection between a computer and the original Ethernet coaxial cable requires a hardware device called a *transceiver*. Physically, the connection between a transceiver and the inner wire of an Ethernet cable enters through a small hole in the outer layers of the cable as Figure 2.2 illustrates. Technicians often use the term *tap* to describe such connections. Usually, small metal pins mounted in the transceiver go through the hole and provide electrical contacts to the center wire and the braided shield. Some manufacturers' connectors require that the cable be cut and a "T" inserted.

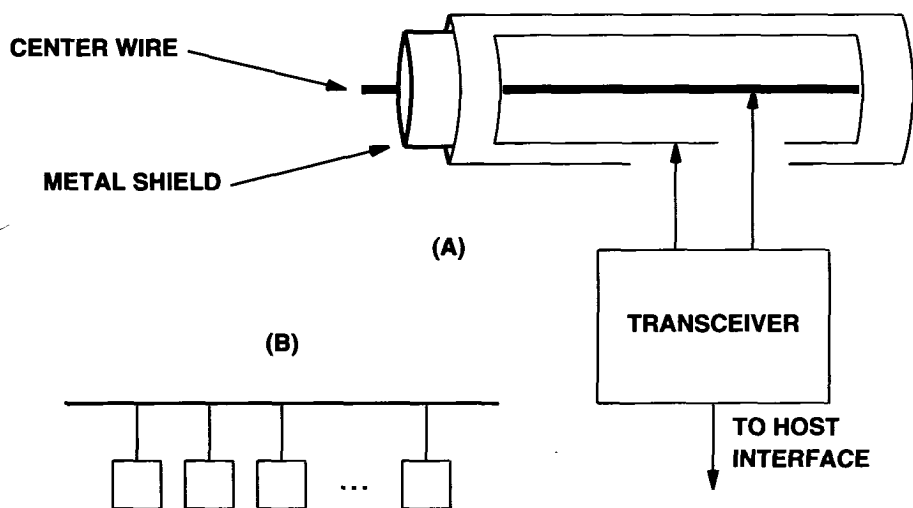


Figure 2.2 (a) A cutaway view of an Ethernet cable showing the details of electrical connections between a transceiver and the cable, and (b) the schematic diagram of an Ethernet with many computers connected.

Each connection to an original Ethernet uses two major electronic components. A *transceiver* connects to the center wire and braided shield on the cable, sensing and sending signals on the ether. A *host interface card* or *host adapter* plugs into the computer's bus (e.g., to a motherboard) and connects to the transceiver.

A transceiver is a small piece of hardware usually found physically adjacent to the ether. In addition to the analog hardware that senses and controls electrical signals on the ether, a transceiver contains digital circuitry that allows it to communicate with a digital computer. The transceiver senses when the ether is in use and translates analog electrical signals on the ether to (and from) digital form. A cable called the *Attachment Unit Interface (AUI)* cable connects the transceiver to an adapter board in a host com-

puter. Informally called a *transceiver cable*, the AUI cable contains many wires. The wires carry the electrical power needed to operate the transceiver, the signals that control the transceiver operation, and the contents of the packets being sent or received. Figure 2.3 illustrates how the components form a connection between a bus in a computer system and an Ethernet cable.

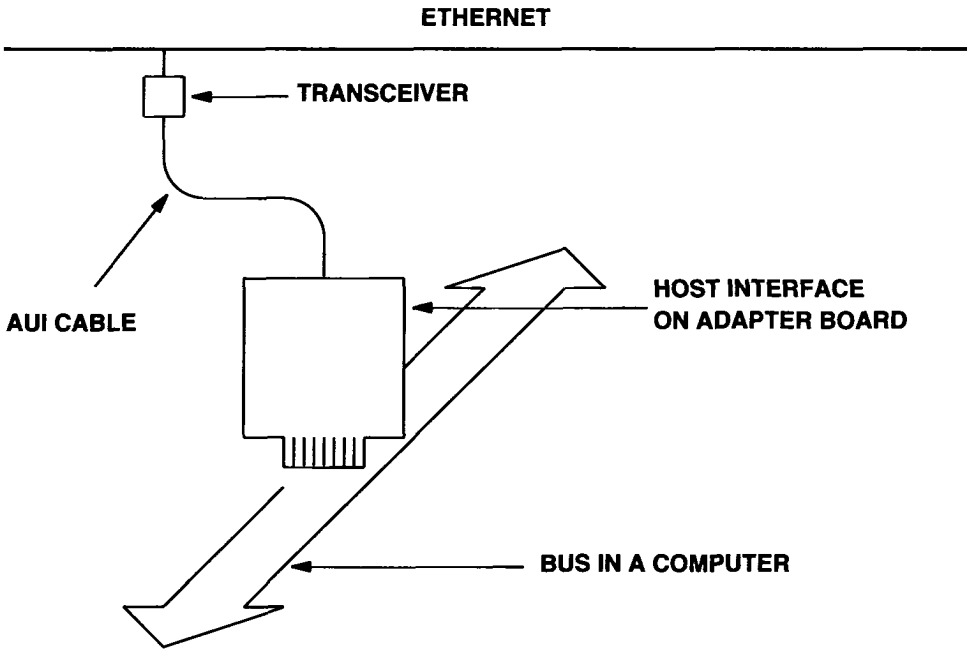


Figure 2.3 The two main electronic components that form a connection between a computer's bus and an Ethernet in the original scheme. The AUI cable that connects the host interface to the transceiver carry power and signals to control transceiver operation as well as packets being transmitted or received.

Each host interface controls the operation of one transceiver according to instructions it receives from the computer software. To the operating system software, the interface appears to be an input/output device that accepts basic data transfer instructions from the computer, controls the transceiver to carry them out, interrupts when the task has been completed, and reports status information. Although a transceiver is a simple hardware device, the host interface can be complex (e.g., some interfaces contain a microprocessor used to control transfers between the computer memory and the ether).

In practice, organizations that use the original Ethernet wiring in a conventional office environment run the Ethernet cable along the ceiling in each hall, and arrange for a connection from each office to attach to the cable. Figure 2.4 illustrates the resulting physical wiring scheme.

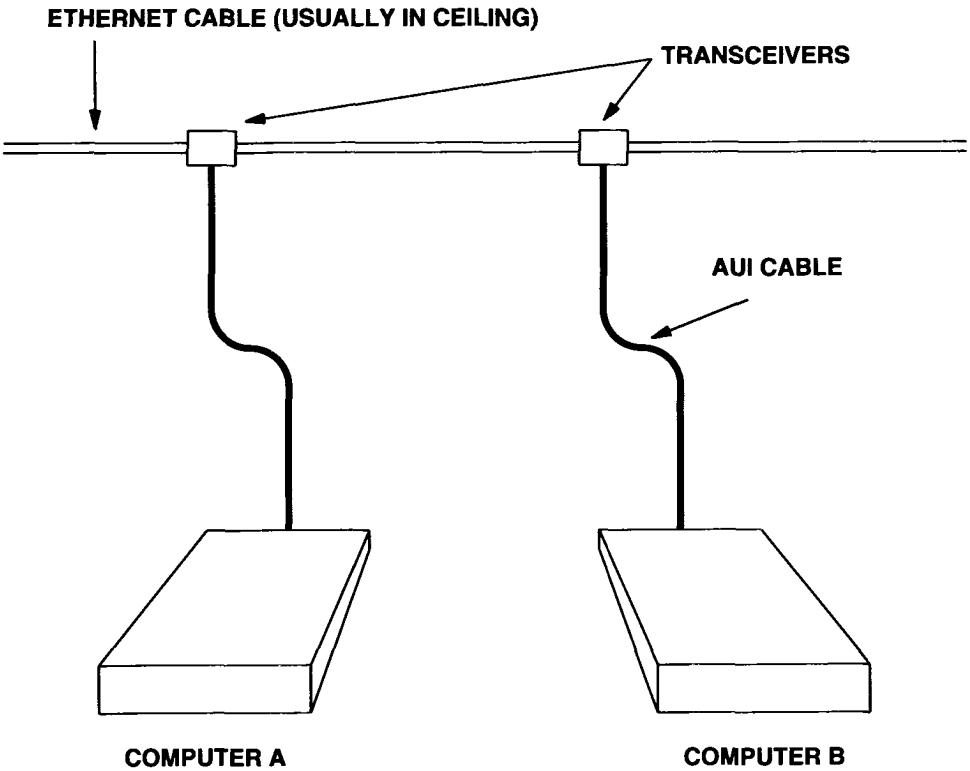


Figure 2.4 The physical connection of two computers to an Ethernet using the original wiring scheme. In an office environment, the Ethernet cable is usually placed in the hallway ceiling; each office has an AUI cable that connects a computer in the office to a transceiver attached to the Ethernet cable.

2.4.1 Thin-Wire Ethernet

Several components of the original Ethernet technology have undesirable properties. For example, because a transceiver contains electronic components, it has a non-trivial cost. Furthermore, because transceivers are located with the cable and not with computers, locating or replacing them is difficult. The coaxial cable that forms the ether is difficult to install. In particular, to provide maximum protection against electrical interference from devices like electric motors, the cable contains heavy shielding that makes it difficult to bend. Finally, the AUI cable is also thick and difficult to bend.

To reduce costs for environments like offices that do not contain much electrical interference, engineers developed an alternative Ethernet wiring scheme. Formally known as *10Base2* and usually called *thin-wire Ethernet* or *thinnet*[†], the alternative coaxial cable is thinner, less expensive, and more flexible. However, thin-wire Ethernet

[†]To contrast it with thin-wire, the original Ethernet cable became known as *thick Ethernet*, or *thicknet*.

has some disadvantages. Because it does not provide as much protection from electrical interference, thin-wire Ethernet cannot be placed adjacent to powerful electrical equipment like that found in a factory. Furthermore, thin-wire Ethernet covers somewhat shorter distances and supports fewer computer connections per network than thick Ethernet.

When designing thin-wire Ethernet, engineers replaced costly transceiver hardware with special high-speed digital circuits, and provided a direct connection from a computer to the network. Thus, in a thin-wire scheme, a computer contains both the host interface and the circuitry that connects to the cable. Manufacturers of small computers and workstations find thin-wire Ethernet an especially attractive scheme because they can integrate Ethernet hardware into single board computers and mount connectors directly on the back of the computer.

Because a thin-wire Ethernet connects directly from one computer to another, the wiring scheme works well when many computers occupy a single room. The thin-wire cable runs directly from one computer to the next. To add a new computer, one only needs to link it into the chain. Figure 2.5 illustrates the connections used with thin-wire Ethernet.

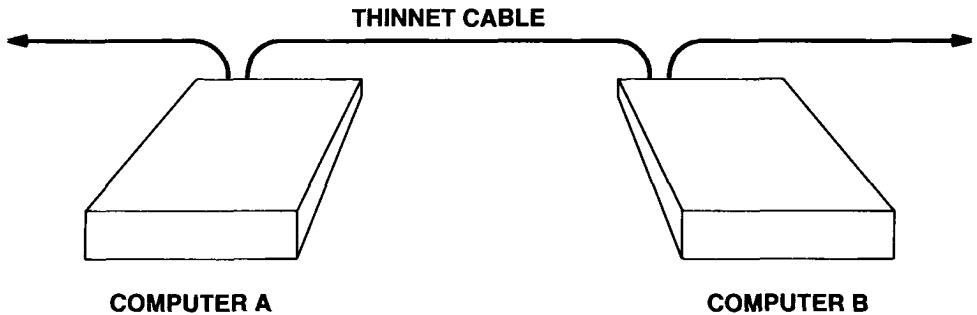


Figure 2.5 The physical connection of two computers using the thinnet wiring scheme. The ether passes directly from one computer to another; no external transceiver hardware is required.

Thin-wire Ethernets were designed to be easy to connect and disconnect. Thin-wire uses *BNC connectors*, which do not require tools to attach a computer to the cable. Thus, a user can connect a computer to a thin-wire Ethernet without the aid of a technician. Of course, allowing users to manipulate the ether has disadvantages: if a user disconnects the ether, it prevents all machines on the ether from communicating. In many situations, however, the advantages outweigh the disadvantages.

2.4.2 Twisted Pair Ethernet

Advances in technology have made it possible to build Ethernets that do not need the electrical shielding of a coaxial cable. Called *twisted pair Ethernet*, the technology allows a computer to access an Ethernet using conventional unshielded copper wires similar to the wires used to connect telephones[†]. The advantages of using twisted pair wiring are that it further reduces costs and protects other computers on the network from a user who disconnects a single computer. In some cases, a twisted pair technology can make it possible for an organization to use Ethernet over existing wiring; in others, the needed wiring (called *category 5 cable*) is cheaper and easier to install than coaxial cable.

Formally known as *10Base-T*, the first twisted pair Ethernet operated at 10 Mbps, exactly like thick or thin Ethernet. A set of eight wires (four pairs) is used to connect each computer to an Ethernet *hub* as Figure 2.6 shows.

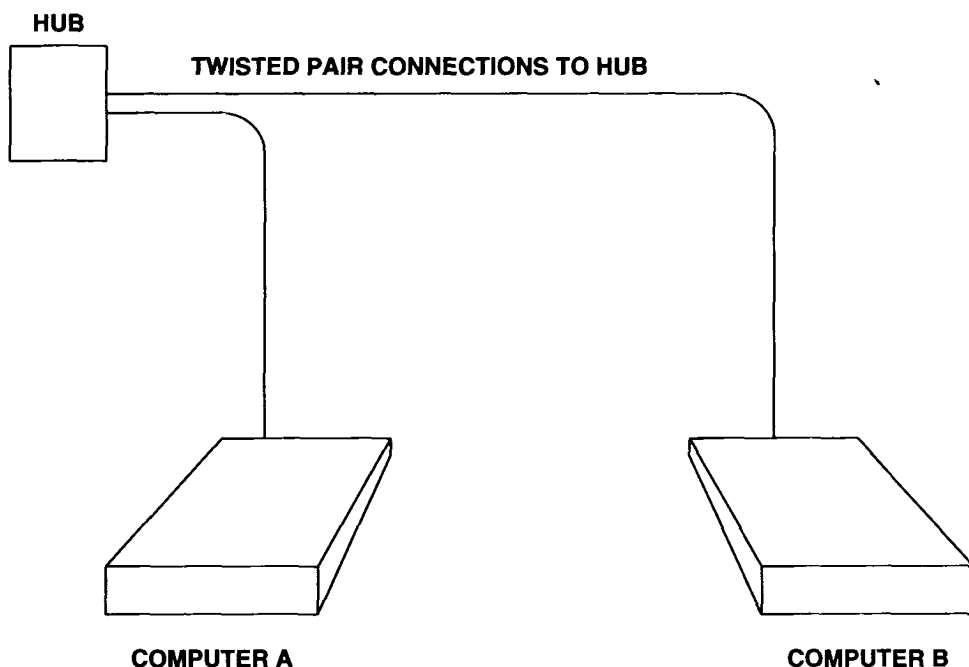


Figure 2.6 An illustration of Ethernet using twisted pair wiring. Each computer connects to a hub over four pairs of wire.

The hub is an electronic device that simulates the signals on an Ethernet cable. Physically, a hub consists of a small box that usually resides in a wiring closet; a connection between a hub and a computer must be less than 100 meters long. A hub requires power, and can allow authorized personnel to monitor and control its operation

[†]The term *twisted pair* arises because conventional telephone wiring uses the technique of twisting the wires to avoid interference.

over the network. To the host interface in a computer, a connection to a hub appears to operate the same way as a connection to a transceiver. That is, an Ethernet hub provides the same communication capability as a thick or thin Ethernet; hubs merely offer an alternative wiring scheme.

2.4.3 Ethernet Capacity

Although the wiring scheme evolved from the original thick cable to thin cable and finally to twisted pair, much of the original Ethernet design remained the same. In particular, the initial twisted pair Ethernet design operates at the same rate as the original thick Ethernet, which means that data can be transmitted at 10 million bits per second. Although a computer can generate data at Ethernet speed, raw network speed should not be thought of as the rate at which two computers can exchange data. Instead, network speed should be thought of as a measure of total traffic capacity. Think of a network as a highway connecting multiple cities, and think of packets as cars on the highway. High bandwidth makes it possible to carry heavy traffic loads, while low bandwidth means the highway cannot carry as much traffic. A 10 Mbps Ethernet, for example, can handle a few computers that generate heavy loads, or many computers that generate light loads.

In the late 1970s when Ethernet was standardized, a LAN operating at 10 Mbps had more than sufficient capacity for many computers because the available CPU speeds and network interface hardware prohibited a given computer from transmitting data rapidly. By the mid 1990s, however, CPU speeds had increased dramatically as had the use of networks. Consequently, an Ethernet operating at 10 Mbps did not have sufficient capacity to act as a central corporate backbone for even a moderate sized corporation — Ethernet had become a bottleneck.

2.4.4 Fast Ethernet

To overcome the throughput limitation of Ethernet, engineers designed a new version of Ethernet that operates an order of magnitude faster. Known formally as *100Base-T*, the technology is usually called *Fast Ethernet*. As the formal name implies, Fast Ethernet uses category 5 twisted pair wiring, the same wiring used for 10Base-T. However, through clever use of the wires, Fast Ethernet allows a station to transmit or receive data at 100 Mbps.

To understand the significance of the increase in capacity, it is important to understand two facts. First, although computers have become faster, few computer systems can transmit data at a sustained rate of 100 Mbps. Second, the 100Base-T standard did not change other parts of the Ethernet standard. In particular, the maximum packet size remains the same as for 10Base-T. These two facts imply that Fast Ethernet is not optimized to provide the highest possible throughput between a pair of computers. Instead, the design is optimized to allow more stations and more total traffic.

2.4.5 10/100 Ethernet

Soon after the invention of Fast Ethernet, manufacturers began to build devices that could accept either a 10 or 100 Mbps connection. The technology, which is known as *dual-speed Ethernet* or *10/100 Ethernet*, is available for computer interfaces as well as for hubs. In essence, all 100Base-T hardware interjects extra signals, making it possible for the hardware at one end of a cable to know which hardware type is connected to the other end. In fact, as long as all eight wires connect to the RJ-45 connector, the cabling and connectors used with 10Base-T are compatible with the cable and connectors used for 100Base-T.

Although 10/100 hardware is slightly more expensive than 10Base-T hardware, it has become extremely popular. Dual speed devices are especially helpful during a transition from 10 Mbps technology to 100 Mbps technology. For example, consider a computer that has a 10/100 interface card. If the computer is connected to a 10Base-T hub, the hardware in the card will automatically detect the speed and communicate at 10 Mbps. If the same computer is then unplugged from the 10Base-T hub and connected to a 100Base-T hub, the hardware will automatically detect the new speed and begin transmitting at 100 Mbps. The transition in speed is completely automatic: neither the software nor the hardware needs to be reconfigured.

2.4.6 Gigabit Ethernet

By the late 1990s, as the market share of 100Base-T Ethernet began to grow, it became obvious that there was a demand for even higher capacity Ethernet. Consequently, engineers extended the Ethernet technology to a bit rate of 1 Gbps (gigabits per second). Known as *1000Base-T*, the high throughput rate makes the technology extremely attractive for use in corporate backbone networks, where traffic from many computers passes through the network. The high data rate does have a slight disadvantage — it makes gigabit Ethernet more susceptible to electrical interference. Consequently, wiring that operates well with 10Base-T or even 100Base-T may not work well with 1000Base-T.

Like Fast Ethernet, the design of gigabit Ethernet was optimized for total throughput. The original packet format and maximum packet size were retained, making packets used on 10Base-T, 100Base-T and 1000Base-T networks interchangeable. Consequently, it is possible to collect traffic from ten 100Base-T Ethernets, each running at full speed, and pass the traffic across a single 1000Base-T network.

2.4.7 Properties of an Ethernet

Ethernet was designed to be a shared bus technology that supports broadcast, uses best-effort delivery semantics, and has distributed access control. The topology is called a *shared bus* because all stations connect to a single, shared communication channel; it is called a *broadcast technology* because all stations receive every transmission, making it possible to transmit a packet to all stations at the same time. The